What makes a “world,” anyway? We just wrote down a single quantum state describing a spin, an apparatus, and an environment. What makes us say that it describes two worlds, rather than just one?

One thing you would like to have in a world is that different parts of it can, at least in principle, affect each other. Consider the following “ghost world” scenario (not meant as a true description of reality, just a colorful analogy): when living beings die, they all become ghosts. These ghosts can see and talk to one another, but they cannot see or talk to us, nor can we see or talk to them. They live on a separate Ghost Earth, where they can build ghost houses and go to their ghost jobs. But neither they nor their surroundings can interact with us and the stuff around us in any way. In this case it makes sense to say that the ghosts inhabit a truly separate ghost world, for the fundamental reason that what happens in the ghost world has absolutely no bearing on what happens in our world.

Now apply this criterion to quantum mechanics. We’re not interested in whether the spin and its measuring apparatus can influence each other—they obviously can. What we care about is whether one component of, say, the apparatus wave function (for example, the piece where the dial is pointing to Up) can possibly influence another piece (for example, where it’s pointing to Down). We’ve previously come across a situation just like this, where the wave function influences itself—in the phenomenon of interference from the double-slit experiment. When we passed electrons through two slits without measuring which one they went through, we saw interference bands on the final screen, and attributed them to the cancellation between the contribution to the total probability from each of the two slits. Crucially, we implicitly assumed that the electron didn’t interact and become entangled with anything along its journey; it didn’t decohere.

When instead we did detect which slit the electron went through, the interference bands went away. At the time we attributed this to the fact that a measurement had been performed, collapsing the electron’s wave function at one slit or another. Everett gives us a much more compelling story to tell.

What actually happened was that the electron became entangled with the detector as it moved through the slits, and then the detector quickly became entangled with the environment. The process is precisely analogous to what happened to our spin above, except that we’re measuring whether the electron went through the left slit L or the right slit R: